

A Domain and Sensor-Adaptive Metric Road Width Estimation from Dual Rear Cameras via Transfer Learning

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Abstract

An accurate measurement of road geometry is important for road infrastructure and maintenance, safety analysis and autonomous systems. While image-based approaches offer an economical substitute to LiDAR, these approaches struggle on roads with degraded or absent lane markings, and rely on forward facing cameras which are highly receptive to occlusions. This paper introduces a novel, domain-, and sensor-adaptive framework for metric road width estimation using dual rear-view cameras on operational road (pavement) survey vehicles. We propose a system based on a fine-tuned road segmentation model with trajectory-aware temporal smoothing and multi-view image fusion via calibrated projections. Extensive validation indicates that our system produces dense, stable, and metrically accurate road width profiles that outperform the marking dependent baselines, especially on unmarked roads. Our system bridges vision-based analysis and infrastructure-grade monitoring offering a practical and high-throughput solution for road network-scale pavement assessment without any additional hardware.

Keywords: Road Width Estimation, Multi-view Fusion, Geometric Reconstruction

1 Introduction

A precise road geometry measurement, particularly metric width, is essential for the maintenance and safety of modern transportation systems. While vision-based sensing is a scalable and cost-effective alternative to LiDAR, perspective distortions, occlusions and degraded (or absent) lane markings make road width estimation challenging. Most common approaches to infer road width through lane marking separation fail on unmarked or non-standard roads[Mahlberg et al., 2022]. Approaches using forward-facing systems (e.g dash cams) are also prone to occlusion and are poorly suited for pavement inspection. In contrast, dual rear-view cameras mounted on survey vehicles offer a stable and unobstructed view right behind the vehicle. However, road segmentation deep learning models trained on public benchmarks suffer from a severe domain shift on rear-view images and poorly marked lanes. Furthermore, per-frame processing of camera video streams leads to temporally unstable predictions.

Road boundary detection has evolved from classical gradient-based edge detection and geometric fitting (e.g., Hough Transform) [Zou et al., 2019] to deep learning approaches. While classical methods were enhanced with probabilistic and temporal filtering, they remained brittle in real-world conditions. Learning-based models such as SegNet [Paszke et al., 2016] improved semantic understanding but are predominantly trained on forward-facing urban datasets and degrade on rear-view imagery of marking-deficient rural roads.

Surround-view systems fuse multiple cameras for 360° road-edge detection [Dahal et al., 2021], but are designed for obstacle avoidance rather than metric infrastructure assessment [Yuan et al., 2023, Hecker et al., 2018]. Vision Transformers have recently advanced segmentation, edge detection [Pu et al., 2022], and line extraction

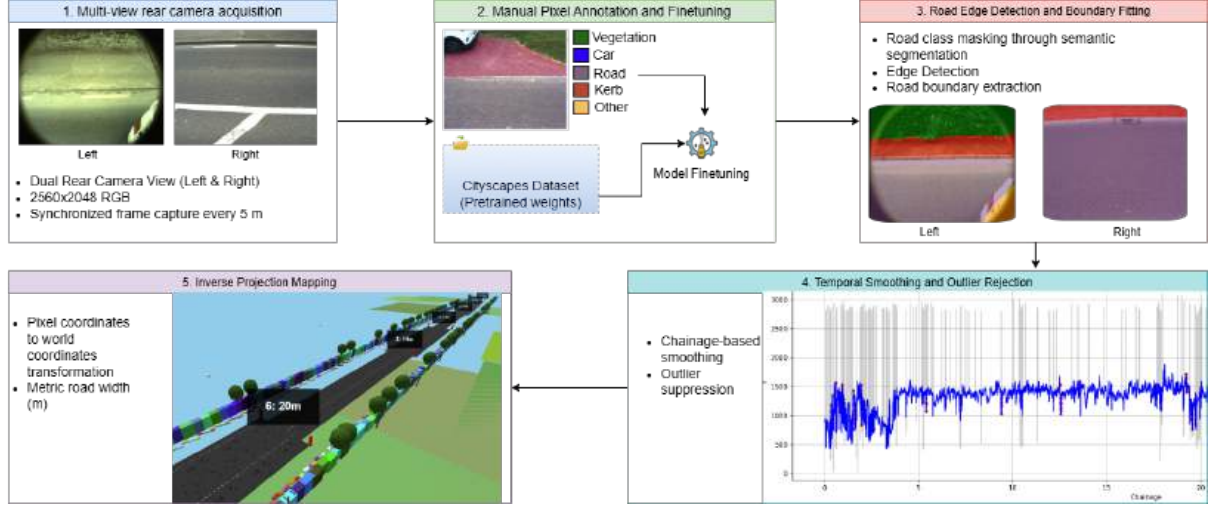


Figure 1: Overview of the proposed methodology for metric road width estimation using dual rear-view cameras. The framework integrates semantic segmentation, geometric boundary estimation, inverse perspective projection, and trajectory-aware temporal refinement.

[Xu et al., 2021], yet models pre-trained on urban data struggle with the appearance and perspective variations of rear-view pavement imagery. Metric width estimation commonly relies on inverse perspective mapping (IPM) with camera calibration and RANSAC fitting [Zou et al., 2019, Ali et al., 2020, Dahal et al., 2021], but assumes clear lane markings [Lu et al., 2025], while temporal models such as SCNNs and ConvLSTM [Dong et al., 2023, Zhang et al., 2021] have been validated primarily on forward-facing video streams.

Our work addresses these limitations through an integrated pipeline that (1) adapts transformer-based segmentation to the rear-view pavement domain, (2) fuses dual rear-camera observations for marking-independent width estimation, and (3) applies trajectory-aware temporal smoothing to produce stable metric profiles. To the best of our knowledge, ours is the first system combining domain-adaptive transformer-based segmentation, dual rear-view fusion, and trajectory-aware estimation for network-level pavement assessment.

2 Methodology

The proposed system operates on an operational pavement survey vehicle equipped with a calibrated dual rear-facing RGB camera rig mounted on a rigid gantry structure, as illustrated in Fig. 2. We propose an end-to-end system for metric road width estimation using synchronized dual rear-view cameras mounted on an operational pavement survey vehicle. The proposed pipeline integrates domain-adaptive semantic segmentation, geometric boundary modeling, metric reconstruction through camera calibration, and trajectory-aware temporal refinement. An overview of our system is illustrated in Fig. 1.

2.1 Sensor Configuration and Data Acquisition

Two cameras are symmetrically positioned on the left and right sides of the vehicle at a fixed height above the road surface and oriented outward with a slight downward pitch, enabling unobstructed observation of pavement edges immediately behind the vehicle. Compared to conventional forward-facing perception systems, this rear-view configuration reduces occlusions from traffic and headlight glare while providing stable post-passage visibility of pavement boundaries. This geometry aligns naturally with *chainage*-

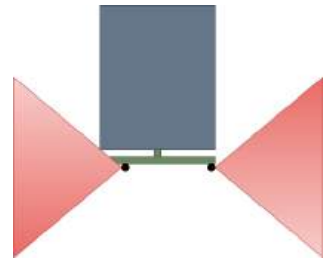


Figure 2: Schematic of the dual rear-view camera configuration mounted on the pavement survey vehicle.

based pavement survey workflows, where measurements are referenced to physical location along the traveled trajectory.

Each camera captures synchronized high-resolution imagery at 2560×2048 pixels, ensuring temporally aligned left–right image pairs for accurate lateral reconstruction. Images are acquired at fixed spatial intervals along the vehicle trajectory, providing consistent coverage for downstream geometric processing. To address the domain gap between publicly available urban datasets and rear-view pavement imagery, a task-specific dataset was constructed from real-world survey dataset covering diverse roads, including urban, rural, and maintenance scenarios. Pixel-level annotations were generated for five semantic categories: road, kerb, vegetation, vehicle, and background.

2.2 Domain-Adaptive Semantic Segmentation

Let $I_t^{(l)}$ and $I_t^{(r)}$ denote synchronized left and right camera images captured at time index t . A semantic segmentation network $f_\theta(\cdot)$ maps each image to dense per-pixel class likelihoods:

$$S_t^{(i)} = f_\theta(I_t^{(i)}), \quad i \in \{l, r\}, \quad (1)$$

$$S_t^{(i)} \in \mathbb{R}^{H \times W \times |\mathcal{C}|} \quad (2)$$

where $|\mathcal{C}|$ represent the total number of semantic classes and the final semantic prediction is obtained as:

$$M_t^{(i)}(x, y) = \arg\max_{c \in \mathcal{C}} S_t^{(i)}(x, y, c). \quad (3)$$

Pixels corresponding to the pavement boundary are extracted by identifying road-class pixels adjacent to a differently-labelled pixel within a local neighbourhood $\mathcal{N}$:

$$E_t^{(i)} = \{(x, y) \mid M_t^{(i)}(x, y) = \text{road}, \exists (x', y') \in \mathcal{N}(x, y) : M_t^{(i)}(x', y') \neq \text{road}\},$$

where jointly considering transitions from road to kerb, vegetation, or other classes keeps the method robust in scenarios where explicit kerb structures are absent, degraded, or partially occluded, while still preserving strong geometric cues for boundary localization.

2.3 Geometric Road Boundary Estimation

For each camera view, the extracted pavement boundary pixels $\mathcal{E}_t^{(i)}$ are approximated using a single straight line in image space. Rather than relying on dense contour extraction, a line-based representation provides a compact and geometrically stable description of road edge, which is suitable for metric projection. A candidate road boundary is represented as

$$\ell_t^{(i)} : a_t^{(i)}x + b_t^{(i)}y + c_t^{(i)} = 0, \quad (4)$$

where x and y denote image coordinates. Due to incorrect decisions caused by shadow, vegetation, or other occlusions, the line parameters are estimated using a robust fitting procedure. This is done by minimizing a robust geometric error over the extracted boundary pixels:

$$\ell_t^{(i)} = \arg\min_{\ell} \sum_{(x_j, y_j) \in \mathcal{E}_t^{(i)}} \rho(|ax_j + by_j + c|), \quad (5)$$

where $\rho(\cdot)$ is a truncated squared loss implemented via RANSAC: residuals below a threshold $\tau = 5$ pixels are treated as inliers and refit by least squares, while larger residuals are excluded, over 3000 trial hypotheses, suppressing outliers arising from shadows, debris, vegetation intrusion, and transient occlusions. After estimating the boundary line, a single boundary observation is obtained by traversing the fitted line from the bottom of the

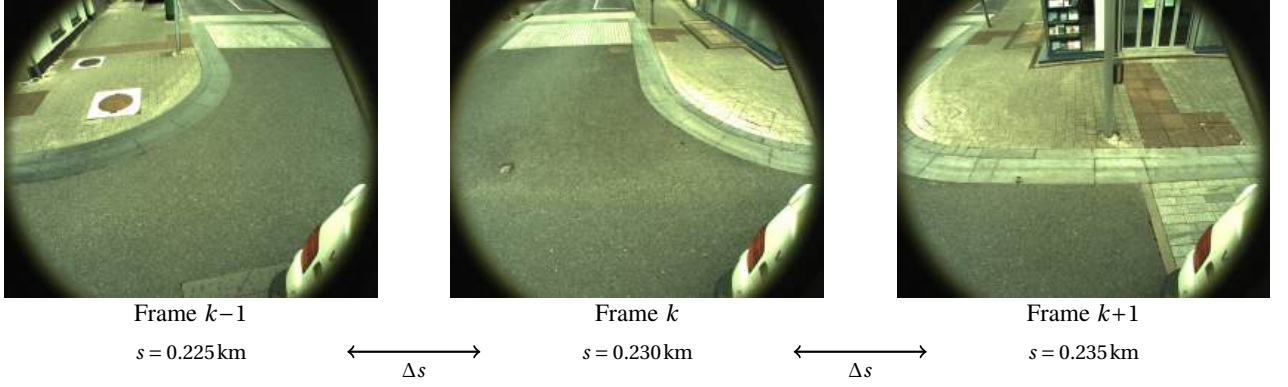


Figure 3: Illustration of trajectory-aware temporal refinement. Rear-view frames are captured at regular spatial intervals (every ~ 5 m) along the vehicle trajectory. For a reference chainage location s_t , width measurements within a spatial neighborhood of length $2\Delta s$ are aggregated, with $\Delta s = 5$ m matching the inter-frame spacing. This chainage-based formulation enables physically meaningful smoothing that is invariant to vehicle speed and frame sampling rate.

image upward and selecting the first row with sufficient local inlier support. Let H denote the image height, and let $x(y)$ denote the fitted line’s x -coordinate at row y :

$$x(y) = -\frac{b_t^{(i)}y + c_t^{(i)}}{a_t^{(i)}}. \quad (6)$$

Starting from $y = H$ and stepping upward, the fitted line is evaluated row by row until a row with sufficient nearby boundary support is found, yielding a single point $p_t^{(i)}$ as the image-space input for metric projection. The search begins at $y = H$ and moves upward because pixels farther from the camera correspond to larger, less-resolved ground patches under inverse perspective projection, so rows closest to the camera are preferred whenever reliable; requiring local inlier support before accepting a row makes the selection robust to localized occlusion or ambiguity, deferring upward only when necessary.

2.4 Metric Reconstruction via Inverse Perspective Projection

Using intrinsic and extrinsic camera calibration, the image-space boundary points $p_t^{(i)}$ obtained in Section 2.3 are projected onto the road surface under a locally planar assumption, with $p_t^{(i)} = (u_t^{(i)}, v_t^{(i)})$. Let $\mathbf{H} = \mathbf{K}[\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{t}]$ denote the ground-plane homography, where $\mathbf{K}$ is the intrinsic matrix and $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{t})$ are obtained from extrinsic calibration.

For each trajectory index t , the left and right image-space boundary observations, $(u_t^{(l)}, v_t^{(l)})$ and $(u_t^{(r)}, v_t^{(r)})$, are mapped to the ground plane using the inverse homography. The metric road width is then computed as the Euclidean distance between the projected boundary points:

$$W_t = \left\| \pi\left(\mathbf{H}^{-1}[u_t^{(l)}, v_t^{(l)}, 1]^\top\right) - \pi\left(\mathbf{H}^{-1}[u_t^{(r)}, v_t^{(r)}, 1]^\top\right) \right\|_2. \quad (7)$$

where $\pi(\cdot)$ denotes normalization of homogeneous coordinates. This produces a sequence of raw road-width estimates along the vehicle trajectory.

2.5 Trajectory-Aware Temporal Refinement

During pavement surveys, images are acquired at fixed spatial intervals along the trajectory. In this configuration, synchronized rear-view frames are captured approximately every 5 m of travel, so consecutive frames provide overlapping views of the same road segment. This introduces a key prior: pavement width varies smoothly

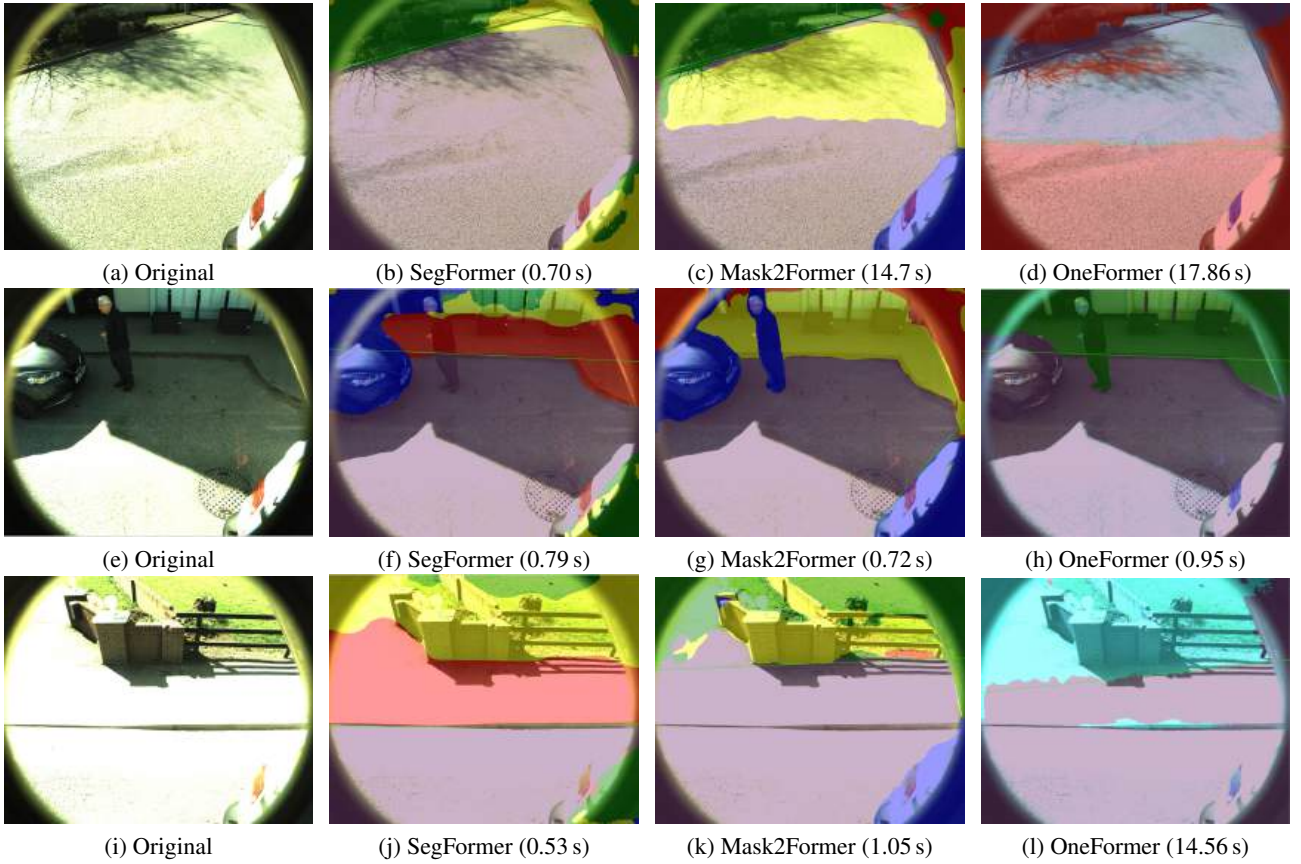


Figure 4: Qualitative comparison of semantic segmentation models with their inference time in seconds. Rows correspond to left rear-view pavement scenes, while columns show the original image and predictions from SegFormer, Mask2Former, and OneFormer. Green lines indicate RANSAC-fitted pavement boundaries fitted to road edge pixels, and red circles denote observation points for geometric evaluation. Quantitatively, SegFormer achieves the highest performance on the rear-view pavement dataset (mIoU: 0.957, mDice: 0.978, Road IoU: 0.995), substantially outperforming Mask2Former (mIoU: 0.465, mDice: 0.581, Road IoU: 0.872) and OneFormer (mIoU: 0.013, mDice: 0.025, Road IoU: 0.051).

along distance and does not change abruptly over short spatial extents under normal conditions. Therefore, neighboring frames contain overlapping geometric information that can improve robustness.

Let s_t denote the chainage (cumulative travel distance) for frame t . Since measurements are referenced to physical location rather than time, refinement is performed in the trajectory domain, ensuring invariance to vehicle speed, frame rate variations, and stop–start motion.

Instantaneous width estimates are susceptible to segmentation uncertainty, geometric amplification during inverse projection, and localized disturbances (e.g., junctions, occlusions). To enforce spatial consistency, width measurements are aggregated within a chainage-aligned neighborhood.

$$\mathcal{W}_t = \{W_k \mid |s_k - s_t| \leq \Delta s\}, \quad (8)$$

where Δs defines the spatial window length. This formulation is illustrated in Fig. 3. Unlike conventional temporal filtering based on frame order, this method aggregates measurements over a fixed physical distance, making refinement directly interpretable for infrastructure analysis. Since neighboring frames are captured every 5 m, $\mathcal{W}_t$ represents multiple observations of the same pavement geometry.

A robust width estimate is obtained by minimizing the ℓ_1 deviation within the local neighborhood.

$$\hat{W}_t = \underset{w}{\operatorname{argmin}} \sum_{W_k \in \mathcal{W}_t} |W_k - w|, \quad (9)$$

The final width is computed by aggregating width estimates within a local chainage neighborhood. This trajectory-aware refinement suppresses isolated outliers while preserving genuine geometric variations, producing smooth, physically consistent width profiles suitable for chainage-based pavement assessment.

3 Experiments and Results

3.1 Segmentation Model Evaluation

As the first and most critical stage of our pipeline, semantic segmentation accuracy directly impacts downstream geometric fitting and temporal aggregation. Therefore, we evaluated different models not only on pixel-level metrics, but also on boundary continuity, geometric stability, and computational efficiency, all essential for operational deployment. We compared three state-of-the-art models: SegFormer [Xie et al., 2021], Mask2Former [Zhang et al., 2022], and OneFormer [Jain et al., 2023], on 201 manually annotated rear-view images (102 left, 99 right) spanning five classes (road, vegetation, kerb, vehicle, background), split 80/20 into training and test sets. Fine-tuned from Cityscapes-pretrained weights for 50 epochs; all models were evaluated on the test set under identical conditions on the same GPU without any test-time augmentation.

Fig. 4 shows qualitative results with RANSAC-fitted boundaries (green lines). SegFormer consistently produces smooth, coherent road masks with well-defined edges, giving a stable boundary even under shadows, occlusions, and perspective distortion. In contrast, Mask2Former gives fragmented masks near boundaries, leading to discontinuous edge points and unstable RANSAC fits. OneFormer suffers from severe class confusion, preventing meaningful boundary extraction quite often. Based on these model evaluations, we selected SegFormer for our segmentation.

3.2 Road-Width Estimation and Results

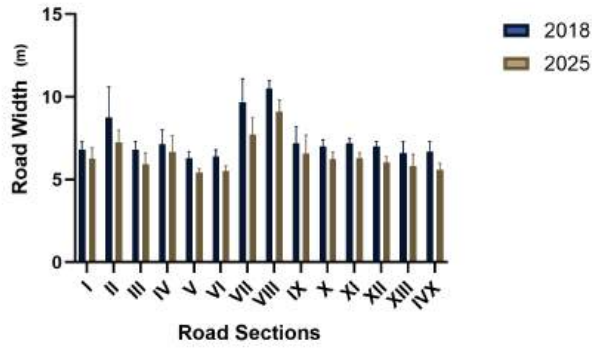
Following model selection, the performance of our road width estimation pipeline is evaluated through a large-scale comparison between historical data collected in 2018 and recent measurements in 2025. In 2018, a single road width was recorded for each 100 m sample unit, whereas in 2025, twenty estimates of rear-view image pairs are acquired per 100 m interval at regular spacing along the vehicle trajectory. Due to this substantial difference in acquisition protocol, direct point-to-point comparison is not meaningful. Instead, we observe geometric consistency, spatial behavior, and network-level agreement using aggregated statistics for each section.

Fig. 5a shows a comparison of historical road widths (2018) vs. estimated widths by our system. It can be observed that the estimated widths are consistently lower than historical measurements while preserving the relative ordering between road sections, indicating that the framework captures meaningful geometric variation rather than converging to a nominal width.

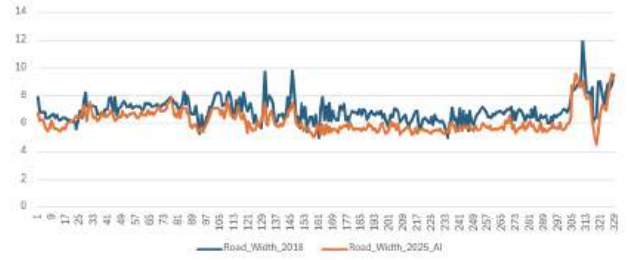
This systematic offset likely reflects a combination of factors: the 2018 and 2025 measurements were captured using different camera systems with independent orientations and calibrations, so any bias in mounting height, pitch angle, or intrinsic parameters propagates directly into the projected width without a shared reference to cross-check consistency; and the seven-year interval between surveys leaves open the possibility of genuine physical change in pavement geometry over time. The consistency and directionality of the offset across sections suggests a systematic rather than random cause, which we intend to isolate through cross-validation against high-accuracy ground truth in future work.

Beyond aggregated averages, the dense measurements enable continuous longitudinal analysis along chainage. As illustrated in Fig. 5b, the estimated width profile closely follows historical trend while exhibiting smooth and stable behavior, which is essential for pavement management systems that rely on consistent chainage based indicators for segmentation and maintenance planning.

This strong alignment empirically validates the trajectory-aware temporal refinement strategy, which aggregates measurements over fixed physical distances rather than frame indices to produce geometrically consistent width estimates. Overall, our system delivers stable, infrastructure-relevant road-width measurements across a



(a) Average road width by section.



(b) Longitudinal road-width profile.

Figure 5: Comparison of 2018 pavement survey widths and 2025 system estimates. (a) Average road widths by representative section, with error bars denoting the standard deviation of chainage-level measurements within each section. (b) Longitudinal width profile along a representative road section, showing the estimated measurements closely following the historical trend while exhibiting smooth and stable behaviour.

diverse regional road network. Although absolute differences relative to historical surveys are observed, these remain systematic and consistent.

4 Conclusion and Future Work

This paper presented a domain-adaptive framework for metric road width estimation using dual rear-view cameras mounted on operational pavement survey vehicles. The proposed framework overcomes key limitations of marking-dependent and forward-facing vision approaches, by combining fine-tuned transformer-based semantic segmentation, calibrated multi-view geometric reconstruction, and trajectory-aware temporal smoothing. Extensive evaluation on real visual survey data demonstrates that the framework produces dense, stable, and metrically consistent width profiles, including on challenging unmarked and rural roads. Our approach requires no additional hardware, integrates seamlessly with existing visual survey workflows, and delivers infrastructure-grade geometric measurements at network scale.

While the proposed framework demonstrates robust performance, it operates under certain practical constraints. First, the 2018 baseline measurements were derived from manually segmented road boundaries, followed by automated metric reconstruction to estimate road width. Consequently, our system’s validation assesses consistency between a semi-automated baseline and a fully automated system, rather than direct manual ground-truth verification. Second, the system assumes the availability of a calibrated dual rear-camera rig, which may not always be available. Third, although trajectory-aware smoothing improves temporal stability, the use of a fixed spatial window (Δs) may not adapt optimally to roads with highly variable geometry or frequent transitions, such as urban intersections or work zones.

Future work will focus on cross-validating vision-based width estimates against high-accuracy manual surveys and mobile LiDAR measurements on selected road sections to establish absolute accuracy bounds. We also plan to incorporate explicit junction and road-transition detection, as well as shadow-aware and illumination-invariant segmentation to improve performance under challenging conditions. Finally, we will investigate adaptive smoothing strategies in which the spatial window (Δs) varies according to road type, curvature, and contextual cues.

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